

JenAlyse M. Arena

Data Analyst · Data Scientist · Statistician

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Data analyst with 14 years of experience in analytics, programming, data visualization, data governance and data pipeline automation. Confident project lead for technology modernization conversions and data-driven research in energy, finance, food safety, and economic sectors. Capable presenter of complex statistical concepts to technical and non-technical stakeholders. Expert technical skills in Python, SAS, SQL, Tableau, PowerBI, and many others.

KEY CAREER ACCOMPLISHMENTS

- Developed multiple effective data visualizations, shepherded agency first public Tableau visualization, authored and automated numerous SAS and python codes, developed multiple master data tables, and two new public datasets.
- Awarded the USDA Administrator's Award for Excellence for 5 years for supporting multiple system migrations and developing innovative dashboards. Awarded the 2023 Under Secretary's Award. Awarded the 2019 FSIS Administrator's Employee of the Year Award. Awarded a Quality Step Increase twice for maintaining data systems and data quality through multiple system infrastructure conversions.

TECHNICAL SKILLS

Python, SAS, SQL, R, Tableau, Power BI, HTML, VBA, Java, Code editors (Jupyter Notebook, Visual Studio Code, Spyder), Query tools (Toad, Aqua Data Studio, SQL Server Mgmt Studio, DBeaver), Text editors (UltraEdit, Notepad++), Oracle, Sybase IQ, SQL Server, Polybase, Azure cloud, Azure Databricks, ArcGIS, Google Earth Pro, SPSS, AWStats, JMP, Minitab, Matlab, Maple, Microsoft tools (Power Platform, Power Automate, Forms, Visio, Project, Excel, Access)

PROFESSIONAL WORK EXPERIENCE

Financial Analyst, Federal Deposit Insurance Corporation

12/2024 –Current

- Designed interactive PowerBI dashboard in two days with optimized user-experience and user-interface demonstrating versatility and speed to create visual analytic insight tools. Leadership was impressed with the capabilities and speed to create KPI and trend monitoring visuals.
- Analyzed complex data parsing code to covert unstructured credit bureau rating data to machine-readable data. The tool replaces manual review of pdf data with automated analysis.
- Lead SAS to Python migration, providing 10+ training resources, tipsheets, and individualized training. Within two months, 80% of staff began Python transition and project was three months ahead of schedule.
- Develop data ingest, transformation, storage, and pipeline solutions including targeted API calls.
- Communicate with system architects, developers, and subject matter experts to develop and implement data integration strategic plans ensuring efficient and accessible data storage, analysis, and dissemination.
- Collaborate across team and agency to align technology goals. Proactively implement knowledge sharing and coding best practice discussions to break siloed work streams.

Technical Data Analyst, Federal Deposit Insurance Corporation

05/2024 –12/2024

- Analyze financial data in SAS, Python, and Azure Databricks. Use new software to compare data transformation model to legacy SAS logic. Programmatically verify data. Identified missing logic and upstream data quality issues.
- Perform extensive user acceptance testing of new public data applications. Verify system requirements match tool functionality. Identify system enhancements to improve user interface and user experience. Within 3 weeks, identified 27 bugs and proposed 4 enhancements.
- Reviewed wire frame system enhancements for optimal user experience within agile project management.
- Provide actionable and constructive technical data solutions to emerging risks. Analyze new financial data tables, identified missing variables, empty tables, and tables with double the expected row count.
- Proactively suggest improvements to communication strategies to ensure cross team collaboration. Confidently and professionally present unresolved bugs to developers. Within 3 months, promoted comprehensive data flow diagram and developed software setup and data access guide to expedite data analysis.

Senior Data Analyst, Food Safety and Inspection Service, USDA**01/2017 – 05/2024**

- Proficient programmer in SAS, R, and Python to perform complex SQL queries, data analyses, statistical tests, and report development from multiple disparate databases.
- Manage 50+ automated codes. Led team through database, software, and platform upgrade in tight 6 and 12-month deadlines. Converted Sybase, SQL, and Polybase to Azure cloud; SAS to Python; and Windows to Linux. Modified code through each conversion maintaining reliable accurate data to 3,000+ field staff and leadership.
- Develop data pipelines for public use files and master datasets. Maintain daily automated refresh code ensuring critical data are readily available. Identify and resolve data quality issues early in development.
- Bridge the communication gap between non-technical customers and technical experts promoting collaboration and inclusion of all data users while providing technical support of mission critical systems. Lead communication between data analysts, data consumers, database administrators, network security, and web application developers to interpret specifications for enhancements and resolve critical system deficiencies.
- Lead Tableau and Google Earth Pro data viz expert. Deploy numerous data visualizations, migrate Tableau servers, and publish first public Tableau dashboard. Serve as Tableau Site Administrator and collaborate with Department on Data Analytics Platform. Build Google Earth Pro mapping tool to identify optimal driving route. Provide regional leaders district reorganization options with coverage and resource impact estimates.
- Manage FSIS email alert system sending messages to over 3,000 federal health inspectors. Promote code to live production server. Perform system testing and provide early go/no-go decision on system deployments.
- Present statistical analyses to agency leaders on public health research. Effectively communicates complex statistical concepts to non-technical audiences demonstrating statistical evidence for public health risk mitigation activities.
- Appointed to detail as Director of the Data Analysis Staff. Direct oversight of two branches, responsible for developing integrated analysis of food safety data to inform policy, including sampling activities, public data postings, and data governance and stewardship for the agency.

Mathematical Statistician, EIA, Dept. of Energy**09/2014 – 01/2017**

- Programmed multiple efficient SAS programs to perform data manipulation, quality assurance, data comparisons, imputation and statistical evaluation from multiple data sources reliably.
- Headed implementation of streamlined coal data transfer process through multiple databases by ensuring compatible table structures and accurate data flow through a reliable and repeatable process.
- Streamlined economic indicator dataset review with a custom SAS program, enabling analysts to validate date and review outliers. Replace existing manual review saving hours of staff time.
- Developed SAS program to automate comprehensive review of key economic indicator dataset allowing analysts to validate data ahead of programmatic enhancement, replacing an existing system of manual review, increasing efficiency and saving hours of staff time.
- Manage operations for coal survey collection, data review, publication, frame maintenance, system updates, form change research and OMB approval, and created quality assurance edit logic.
- Authored EIA Today in Energy coal-related articles that were well received by market analysts.
- Utilize multiple data systems and SQL to respond to data inquiries from federal agencies, policy-makers at the federal and state levels, reporters, academic institutions, respondents, and the general public.
- Received Innovation Award in 2016 for collaborative work to automate and redesign coal markets webpage which reduced manual processes and provided more relevant key market indicators.

Survey Statistician, U.S. Census Bureau**06/2011 – 09/2014**

- Conducted data collection, compilation, statistical review, and dissemination of public pension data.
- Authored technical documents for survey forms, mail-out, data collection, processing, and table specifications.
- Designed and implemented customized Microsoft Access database replacing costly licensed software.
- Analyzed agency-wide customer satisfaction KPI using survey, web traffic, and call center data.
- Fostered relationships with respondents and data users to resolve inquiries in the agency's main call center.

EDUCATION**Bachelor of Arts in Mathematics and Statistics, St. John Fisher College, Rochester, NY**